MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 10 · ALGEBRA II · WEEK 7

Pre-Session

Lesson 2-4

Overview of Complex Numbers

Lesson 2-5

Completing the Square

Mr Malek Thiab

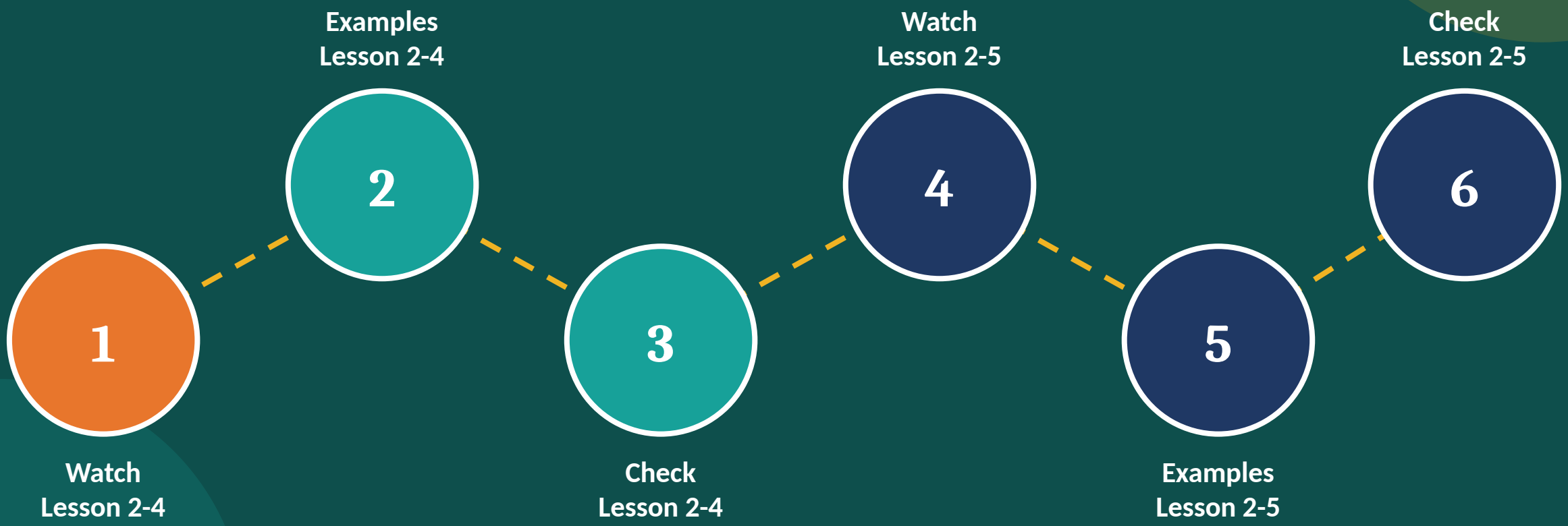
Home task before class

Watch · Study · Check · Submit on the LMS before class

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Your home journey

Six stations before the lesson: watch, study the examples, check yourself, twice.



STATION 1 · WATCH · LESSON 2-4

Overview of Complex Numbers

WATCH BEFORE CLASS · 5:19

Introduction to i and imaginary numbers

Khan Academy · 5:19

Open on YouTube

LISTEN FOR

1. What number the imaginary unit i is defined to be, and what i squared equals.
2. How a square root of a negative number is rewritten using i.
3. How the powers of i repeat in a cycle, and the form a + bi for a complex number.

KEY IDEA

$$i = \sqrt{-1}, \qquad i^2 = -1, \qquad z = a + bi$$

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$

STATION 2 · EXAMPLES · LESSON 2-4

Study the worked examples

EXAMPLE A

1

Simplify: $\sqrt{-48}$

$$= \sqrt{48} \cdot \sqrt{-1}$$

$$= 4\sqrt{3}\,i$$

EXAMPLE B

2

Multiply: $(3 + 2i)(1 - 4i)$

$$= 3 - 12i + 2i - 8i^2$$

$$= 3 - 10i + 8$$

$$= 11 - 10i$$

STATION 3 · CHECK · LESSON 2-4

Check yourself: three questions

1

Simplify $\sqrt{-75}$

2

Evaluate i^{43}

3

Find $(1+2i)+(3-i)$ and $(1+2i)(3-i)$

Done when

you can write a square root of a negative number with i, use the cycle of four to find a power of i, and give sums and products in the form a + bi.

HSN.CN.A.1 · HSN.CN.A.2 · HSN.CN.A.3 · HSN.CN.C.8 · HSA.REI.B.4.B · HSN.CN.C.7 · HSA.SSE.B.3.B · HSA.REI.B.4.A · HSF.LE.B.6 · HSF.IF.C.8.A

STATION 4 · WATCH · LESSON 2-5

Completing the Square

WATCH BEFORE CLASS · 6:14

Completing the Square | From Standard to Vertex Form

Brian McLogan · 6:14

Open on YouTube

LISTEN FOR

1. Why half of the coefficient of x is found and then squared.
2. How the expression stays equal after a number is added to it.
3. How the vertex and the minimum value can be read from the finished form.

KEY IDEA

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$$

$$y = (x - h)^2 + k : \text{ vertex } (h, k), \text{ minimum } k$$

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STATION 5 · EXAMPLES · LESSON 2-5

Study the worked examples

EXAMPLE A

1

Rewrite $x^2 + 6x - 7 = 0$ as $(x + p)^2 = q$

$$x^2 + 6x = 7$$

$$x^2 + 6x + 9 = 7 + 9$$

$$(x + 3)^2 = 16$$

EXAMPLE B

2

Vertex form of $y = x^2 - 8x + 11$

$$y = x^2 - 8x + 16 - 16 + 11$$

$$y = (x - 4)^2 - 5$$

$$\text{vertex } (4, -5),\ \min = -5$$

STATION 6 · CHECK · LESSON 2-5

Check yourself: three questions

1

Write $x^2 + 10x - 2 = 0$ as $(x + p)^2 = q$

2

Write $y = x^2 - 6x + 5$ in vertex form

3

Find the minimum of $y = x^2 + 4x + 9$

Done when

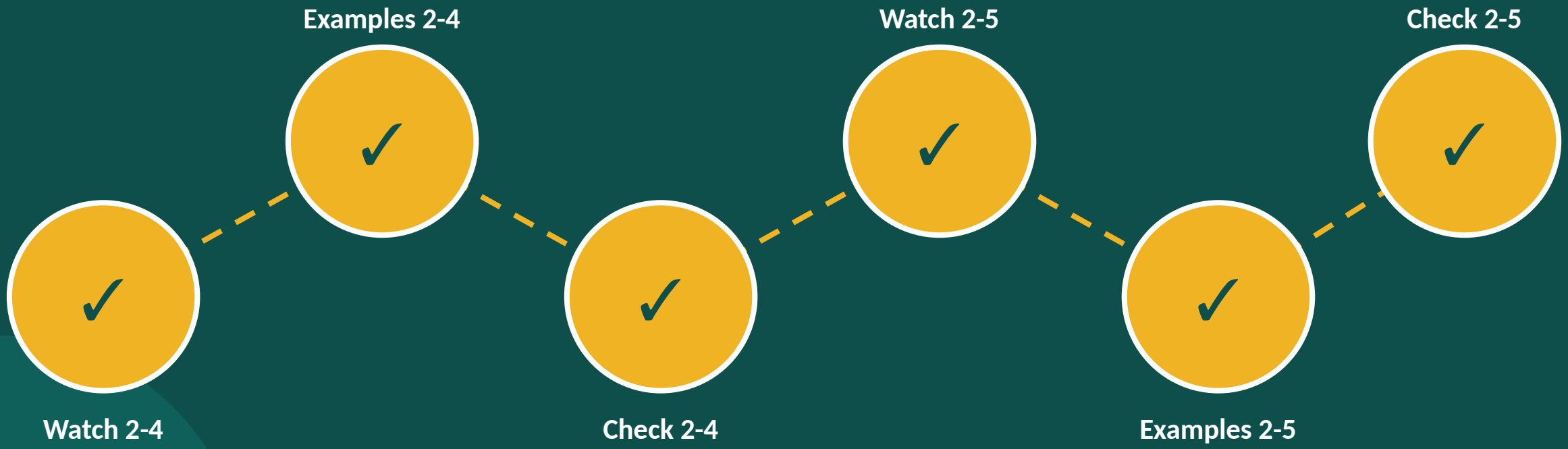
you can add the square of half the coefficient of x to both sides of an equation, and write a quadratic with leading coefficient 1 in vertex form and name its minimum value.

HSN.CN.A.1 · HSN.CN.A.2 · HSN.CN.A.3 · HSN.CN.C.8 · HSA.REI.B.4.B · HSN.CN.C.7 · HSA.SSE.B.3.B · HSA.REI.B.4.A · HSF.LE.B.6 · HSF.IF.C.8.A

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Ready for class

Bring your answers and your questions.



Last time you solved quadratics and met the discriminant; this week a negative discriminant gets a number, and completing the square rewrites a quadratic so its minimum or maximum shows.

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